

Swift Ship Tracker

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SWIFT SHIP TRACKER

USER STORY

The SwiftShip Tracker is designed to provide a centralized Salesforce-based parcel management solution that enables customers to book parcels, track delivery status, and manage shipments through an intuitive interface and conversational AI support. The system manages Parcel, Delivery, Sender, and Receiver information using Salesforce custom objects and relationships. Flows automate parcel operations and notifications, while AgentForce AI and Prompt Builder enable customers to retrieve parcel information and interact with the system conversationally. Delivery agents can update parcel status in real time, while Experience Cloud, security controls, reports, and dashboards provide secure customer access and operational visibility. intha mathiri

ABSTRACT

SwiftShip Tracker is a centralized Salesforce-based parcel management solution that enables customers to book parcels, track delivery status, and manage shipments through an intuitive interface and conversational AI support. The system manages Parcel, Delivery, Sender, and Receiver information using Salesforce custom objects and relationships. Flows automate parcel operations and notifications, while AgentForce AI and Prompt Builder enable customers to retrieve parcel information and interact with the system conversationally. Delivery agents can update parcel status in real time, while Experience Cloud, reports, dashboards, and security controls support customer service and operational management.

The project is organized into requirement analysis, ideation, requirement analysis, design, planning, Salesforce backend development, UI/UX and security configuration, AgentForce implementation, testing, and documentation/maintenance. The implementation evidence supplied for this report includes the Swift Ship Tracker workspace, Salesforce Object Manager, Permission Sets, Flow Builder, Prompt Builder, and Agentforce Studio.

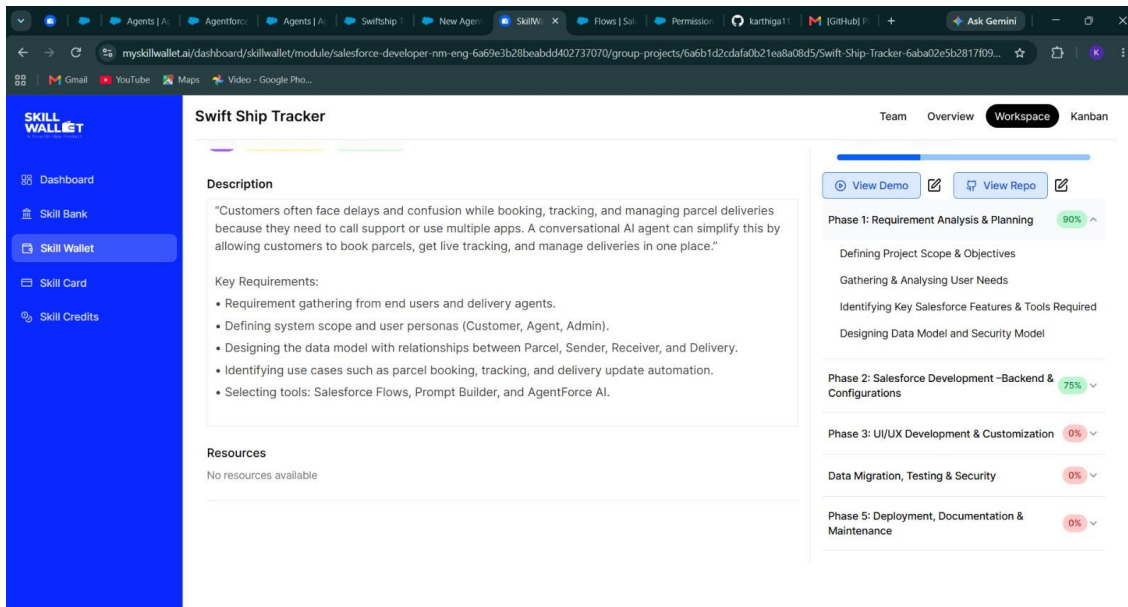


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1. INTRODUCTION

1.1 Project Overview

The SwiftShip Tracker is a Salesforce CRM solution developed to simplify parcel booking, shipment tracking, and delivery management. It supports the complete parcel lifecycle from booking and dispatch through tracking and delivery confirmation. Customers can access shipment information, while delivery agents can update parcel details and delivery status.

1.2 Purpose

- Centralize parcel and delivery information.
- Simplify parcel booking and tracking.
- Provide real-time shipment visibility.
- Automate delivery updates and customer notifications.
- Enable conversational parcel tracking using AgentForce AI and Prompt Builder.
- Improve agent efficiency through Salesforce Flows and dashboards.
- Provide secure, role-based access to customers, agents, support staff, and administrators.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am	I'm trying to	But	Because	Which makes me feel
PS-1	Customer / Sender	book a parcel easily	booking requires support assistance or multiple applications	parcel operations are not centralized	frustrated
PS-2	Customer / Sender	track my parcel	there is no simple real-time tracking experience	shipment information is difficult to access	confused
PS-3	Delivery Agent	update parcel status	status updates require manual handling	centralized operational tracking is needed	concerned
PS-4	Customer	receive delivery updates	notifications may not be available at every stage	communication needs automation	uncertain
PS-5	Admin	monitor parcel operations	parcel flow and delivery performance are difficult to monitor	operational data needs centralized reporting	concerned

2.2 Empathy Map Canvas

- Think & Feel: Wants a simple way to book and track parcels without repeatedly contacting support.
- Hear: Receives delivery information and status updates from the parcel service.
- See: Parcel details, tracking status, estimated delivery date, and delivery information.
- Say & Do: Books parcels, checks tracking information, and asks for delivery updates.
- Pain: Delays, confusion, lack of easy tracking, and dependence on support or multiple applications.
- Gain: Real-time visibility, self-service tracking, convenient booking, and timely notifications.

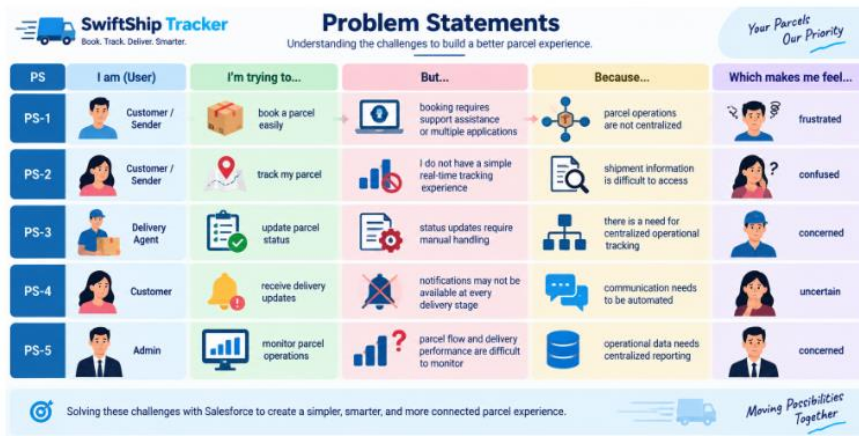


Figure 2.1. Empathy-map/ideation visual reproduced from the supplied reference. 2.2 Empathy Map Canvas

User Persona: Customer / Sender

- Think & Feel: Wants a simple way to book and track parcels without repeatedly contacting support.
- Hear: Receives delivery information and status updates from the parcel service.
- See: Parcel details, tracking status, estimated delivery date, and delivery information.
- Say & Do: Books parcels, checks tracking information, and asks for delivery updates.
- Pain: Delays, confusion, lack of easy tracking, and dependence on support or multiple apps.

2.3 Brainstorming & Idea Prioritization

- Manual registers
- Spreadsheet tracking
- Standalone parcel portal
- Salesforce CRM with AgentForce AI

Final Selection: Salesforce CRM was selected in the reference because it provides centralized parcel management, automation through Flows, real-time tracking visibility, conversational AI assistance, secure role-based access, and reports and dashboards.

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

- Parcel Booking
- Parcel Record Creation
- Dispatch & Shipment
- In-Transit Tracking
- Out for Delivery
- Delivery Confirmation
- Customer Notification
- Feedback / Support

3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Parcel booking and record creation
FR-2	Parcel status tracking
FR-3	Weight management
FR-4	Estimated delivery date management
FR-5	Sender information management
FR-6	Receiver information management
FR-7	Delivery and current-location tracking
FR-8	Agent parcel status updates
FR-9	Automated customer notifications
FR-10	AI-powered parcel tracking
FR-11	Reports and dashboards
FR-12	Customer access through Experience Cloud

Non-Functional Requirements

NFR	Description
Usability	Intuitive Salesforce and customer-facing interface
Security	Role-based and field-level access control
Reliability	Accurate parcel status and delivery information
Performance	Timely Flow and AgentForce responses
Availability	Salesforce cloud-based access
Scalability	Supports growing parcel and delivery operations
Data Integrity	Controlled access and relationship-based parcel data

3.3 Data Flow

Parcel Booking → Parcel Record → Sender + Receiver Information → Delivery Record → Parcel Status Update → Current Location & Estimated Delivery → Automated Notifications → Customer Tracking / AgentForce AI → Reports & Dashboards.

AI Tracking Flow: Customer Query → AgentForce AI → Prompt Builder – Retrieve Parcel Details → Parcel Record → Status / Weight / Estimated Delivery → Conversational Response.

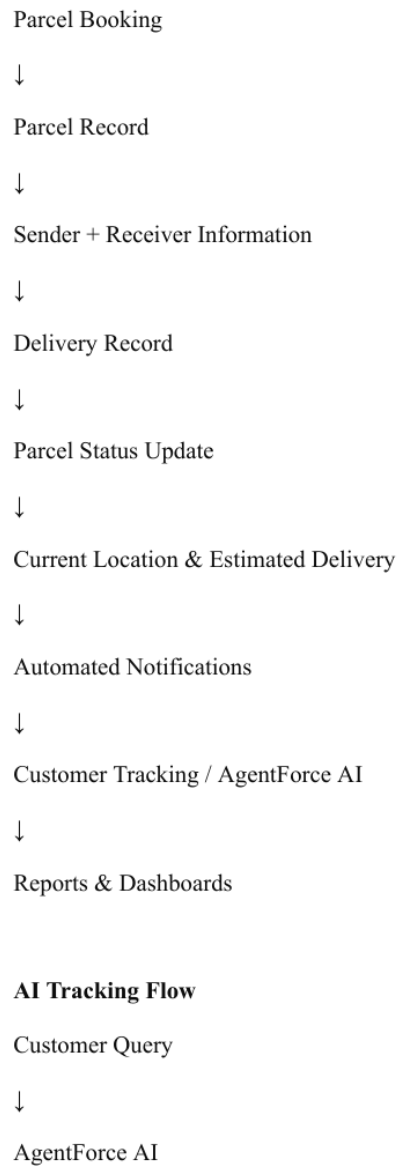


Figure 3.1. Data-flow / AI-tracking visual from the supplied reference.

3.4 Technology Stack

Layer	Technology
Platform	Salesforce Developer Edition
UI	Lightning App Page, Lightning Record Pages, Experience Cloud
Database	Salesforce Custom & Standard Objects
Business Logic	Salesforce Flows, Apex Classes

Automation	Record-Triggered Flows, Auto-Launched Flows, Scheduled Flows
AI	AgentForce AI, Prompt Builder
Notifications	Email Alerts & Email Templates
Processing	Batch Apex
Security	Profiles, Permission Sets, Role Hierarchy, Field-Level Security, Sharing Rules
Reporting	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

Problem	Solution	Benefit
Difficult parcel booking and tracking	Salesforce-based Parcel Management System	Centralized booking and tracking
Delivery communication gaps	Email Alerts	Timely notifications
Scattered shipment information	Parcel, Delivery, Sender & Receiver objects	Structured data
Difficult operational monitoring	Reports & Dashboards	Better visibility
Different users need different access	Profiles, Roles & Permission Sets	Secure access

4.2 Proposed Solution

Parameter | Description

Parameter	Description
Problem	Difficult and inefficient parcel booking and tracking
Solution	Salesforce-based Parcel Management System
Innovation	AgentForce AI + Prompt Builder for conversational self-service tracking and automated Salesforce Flows
Impact	Centralized booking, real-time tracking visibility, timely notifications, and efficient delivery management
Business Model	Parcel, customer, delivery-agent, support, and administrator operations
Scalability	Supports growing parcel operations, Experience Cloud, and integration with external logistics systems

4.3 Solution Architecture

- Users: Customer, Delivery Agent, Support Staff, Admin
- Interface: Salesforce Lightning / Experience Cloud
- Data Layer: Parcel, Delivery, Sender, Receiver
- Automation: Flows, Approval Processes, Apex, Batch Apex, Email Alerts
- AI Layer: AgentForce AI, Prompt Builder, Parcel Tracking Agent
- Security Layer: Profiles, Roles, Permission Sets, Sharing Rules, Field-Level Security
- Reports & Dashboards

Parameter	Description
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4.3 Solution Architecture

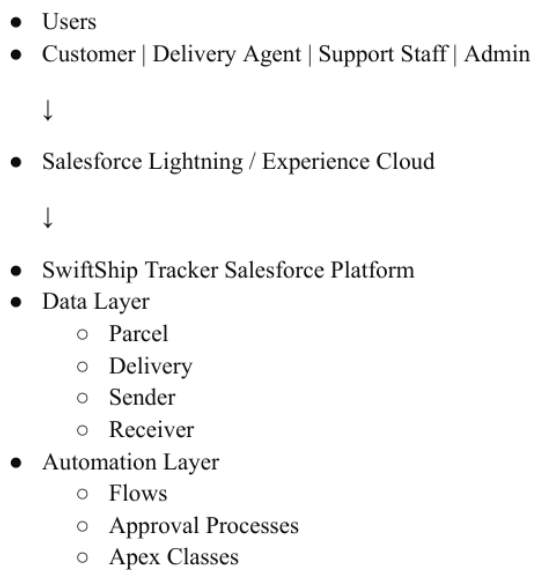


Figure 4.1. Solution architecture visual from the supplied reference.

4.4 Data Model and Security Model

Custom objects: Parcel__c, Delivery__c, Sender__c, Receiver__c.

- Parcel__c: Parcel ID, Status, Weight, Estimated Delivery Date; linked with Sender, Receiver and Delivery.

- Delivery__c: Current Location, Estimated Delivery Date, Sender, Parcel.
- Sender__c: Sender Address, Sender Contact, Sender Email.
- Receiver__c: Receiver Address, Receiver Contact, Receiver Email, Sender, Parcel.
- Role hierarchy: Administrator → Delivery Manager → Agent → Customer.
- Admin Profile: Full access to all objects and records.
- Delivery Agent Profile: Read/Edit access to Parcel, Delivery, Sender, and Receiver.
- Customer Profile: Read-only access to their own parcel details through Experience Cloud.
- Support Staff Profile: Access to Cases and Parcel details for issue resolution.
- Permission Sets provide additional privileges such as Update Delivery Status and Access Agent Console.
- Sharing Rules and Field-Level Security provide controlled record and field visibility.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

Product Backlog, Sprint Schedule, and Estimation

Sprint	Epic	User Story	Task	Points	Priority
Sprint-1	Developer Setup	USN-1	Create and configure Salesforce environment	3	High
Sprint-2	Data Modeling	USN-2	Create Parcel, Delivery, Sender and Receiver objects with relationships	5	High
Sprint-2	Data Modeling	USN-3	Create dedicated tabs and Lightning App	5	High
Sprint-3	Automation	USN-4	Validate parcel details and mandatory fields	3	High
Sprint-3	Automation	USN-5	Automate parcel status and delivery information using Flows	3	High
Sprint-4	Apex	USN-6	Use Apex classes/triggers for critical parcel and delivery actions	5	High
Sprint-4	Apex	USN-7	Use Batch Apex for periodic updates	5	High
Sprint-5	Reports	USN-8	Create reports for parcel status and delivery performance	4	High

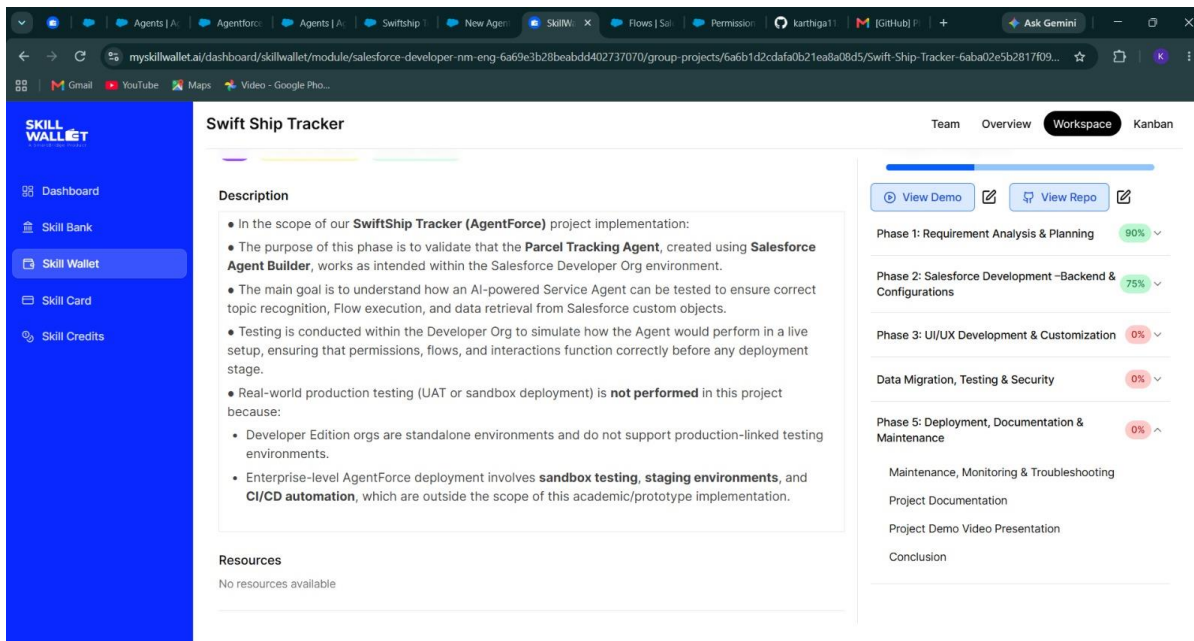
Sprint-6	Dashboards & AI	USN-9	Create dashboards and AgentForce AI tracking assistance	4	Medium
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6. PROJECT DEVELOPMENT & IMPLEMENTATION

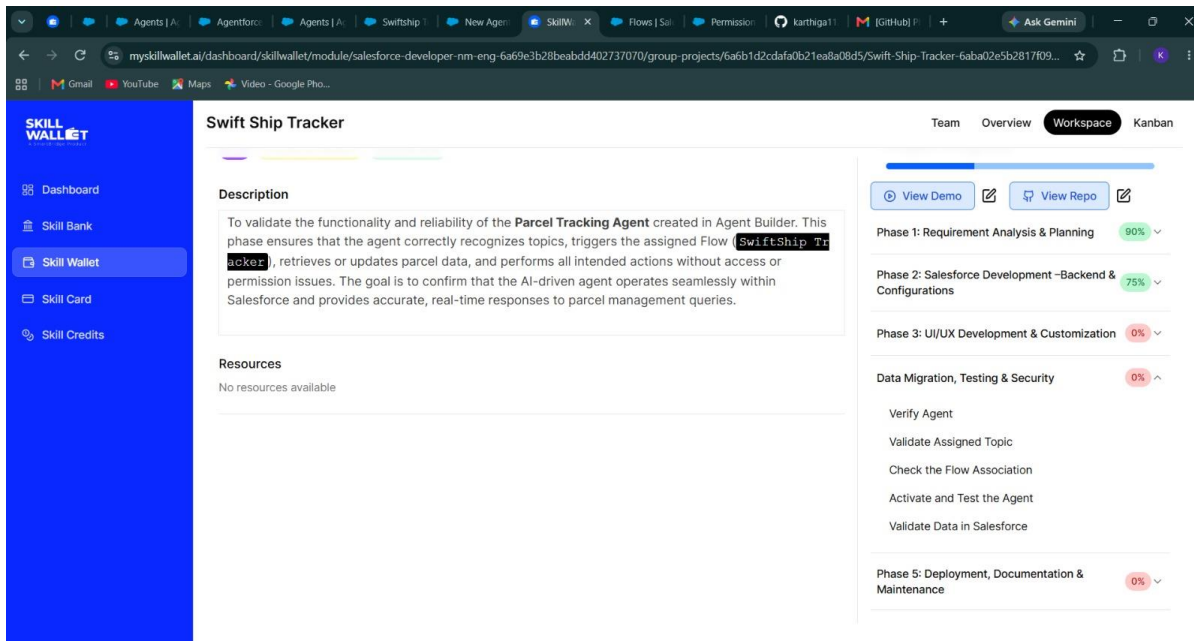
This section follows the supplied reference file's implementation sequence and incorporates the user's Salesforce screenshots as implementation evidence.

6.1 Requirement Analysis & Planning

- Defined project scope and objectives.
- Identified Customer, Delivery Agent, Support Staff and Admin personas.
- Designed the Parcel, Delivery, Sender and Receiver data model.
- Selected Salesforce Flows, Prompt Builder and AgentForce AI.
- Defined security requirements and role-based access.



SkillWallet – Parcel Tracking Agent validation/testing task.



SkillWallet – Phase 5 deployment, documentation and maintenance scope.

6.2 Salesforce Development – Backend & Configurations

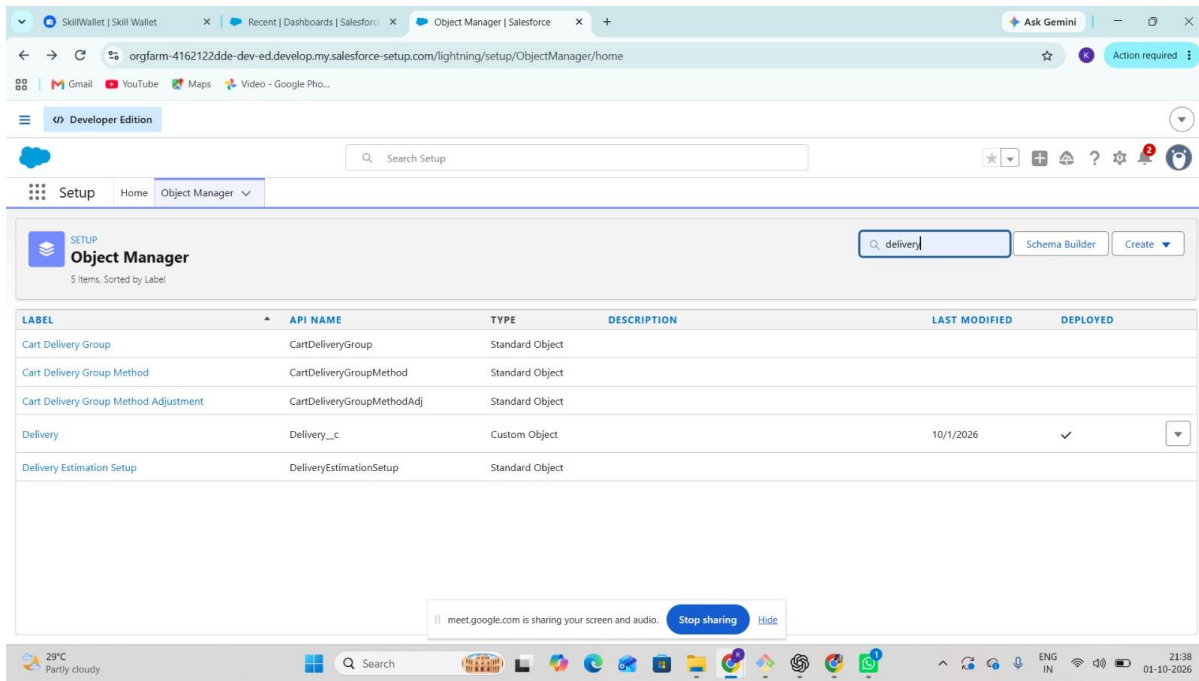
The backend phase creates the core Salesforce objects, fields, relationships, Prompt Builder template and Flow automation.

Milestone 1 – Salesforce Account

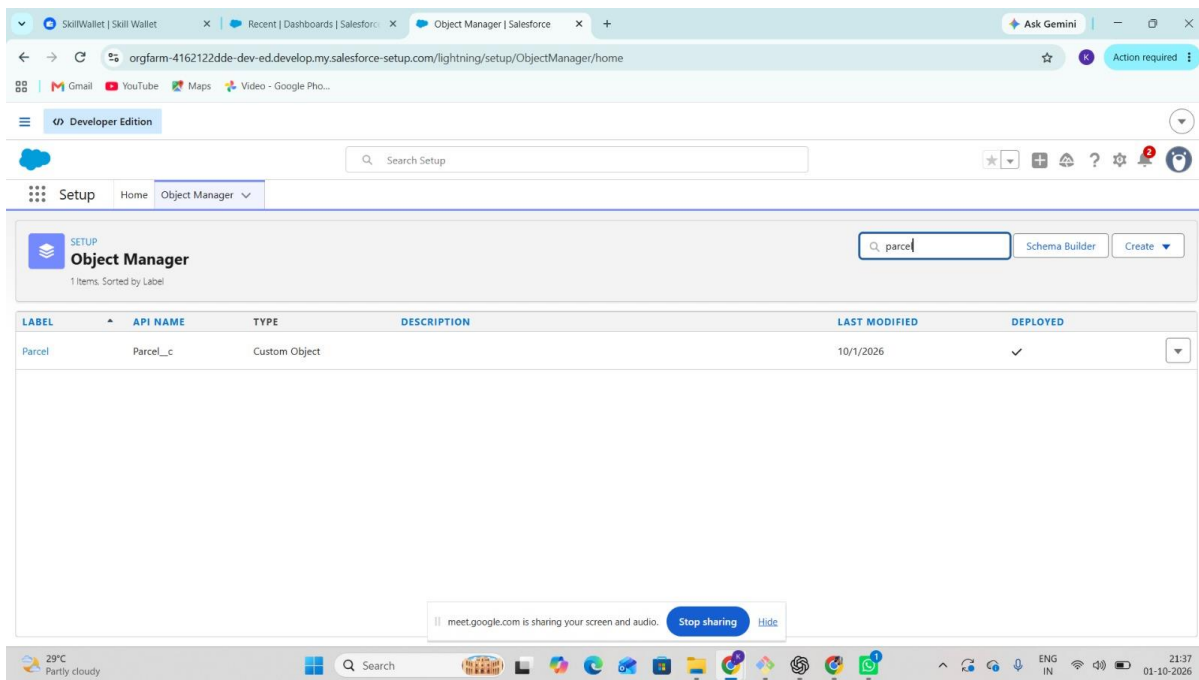
A Salesforce Developer Edition org is used for configuration, development and testing.

Milestone 2 – Objects Creation

- Parcel custom object
- Delivery custom object
- Sender custom object
- Receiver custom object



Salesforce Object Manager – Parcel custom object.



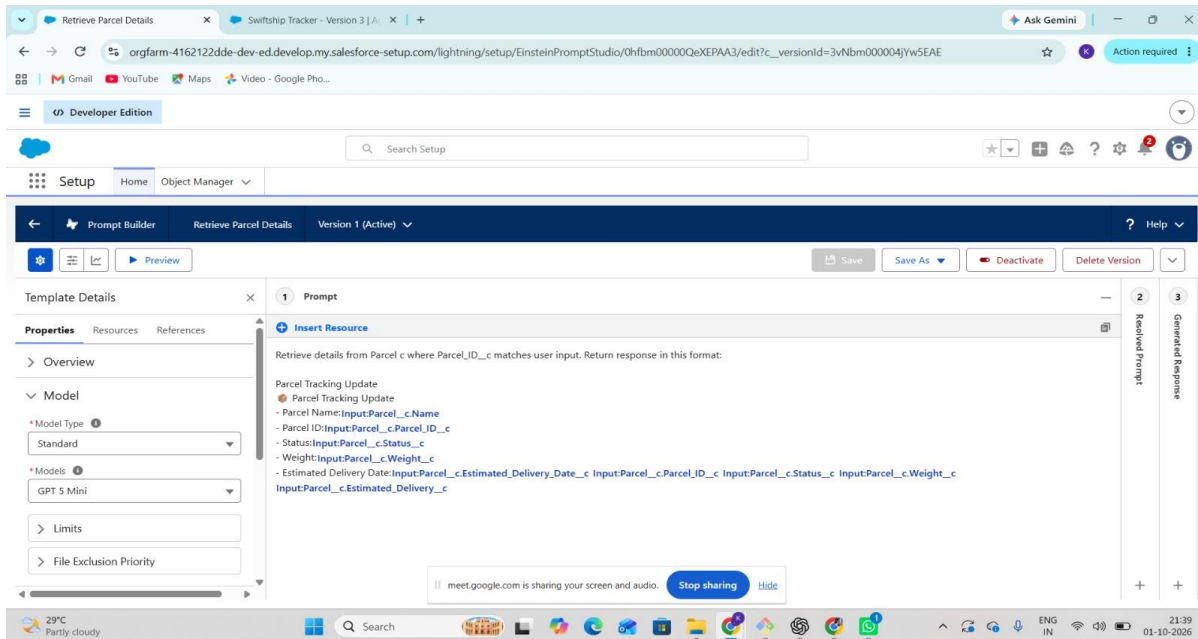
Salesforce Object Manager – Delivery custom object.

Milestone 3 – Tabs

Custom object tabs are created for Parcel, Delivery, Sender and Receiver to support navigation and module access.

Milestone 4 – Fields & Relationships

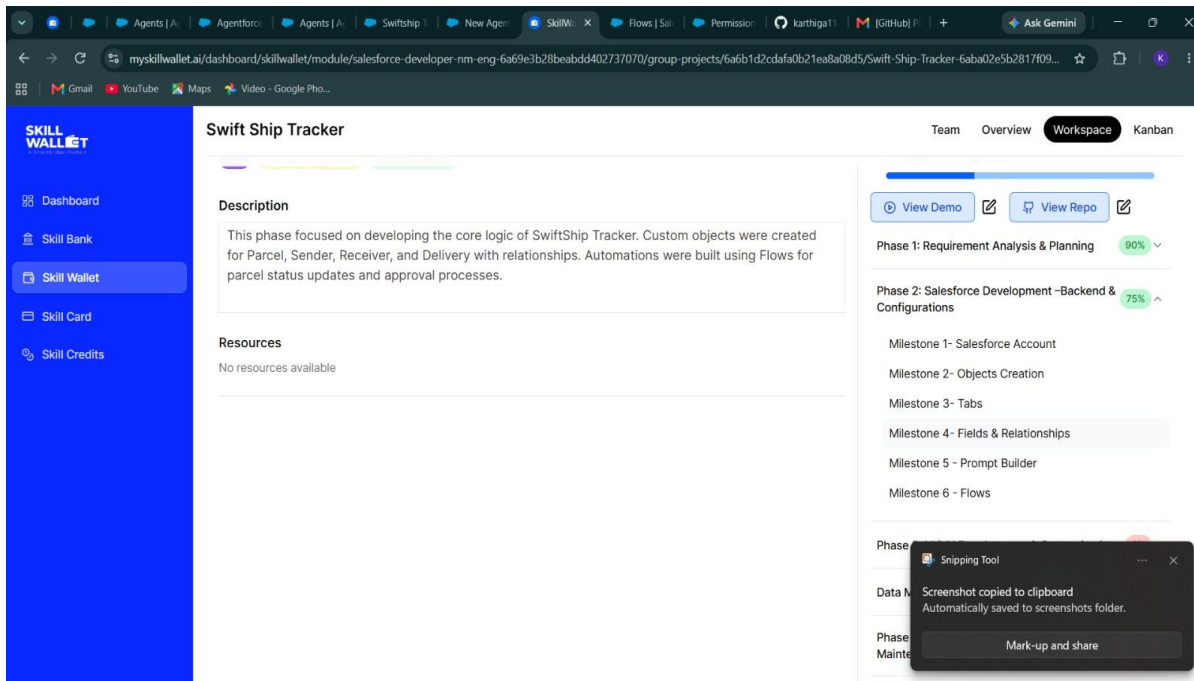
- Parcel: Parcel ID (Auto Number P-{000}), Status (Picklist), Weight (Number), Estimated Delivery (Date), Sender (Lookup).
- Delivery: Current Location (Geolocation), Estimated Delivery (Date), Sender (Lookup), Parcel (Lookup).
- Sender: Sender Address (Geolocation), Sender Contact (Phone), Sender Email (Email).
- Receiver: Receiver Address (Geolocation), Receiver Contact, Receiver Email, Sender, Parcel.



SkillWallet – Phase 2 milestones: Salesforce Account, Objects, Tabs, Fields & Relationships, Prompt Builder and Flows.

Milestone 5 – Prompt Builder

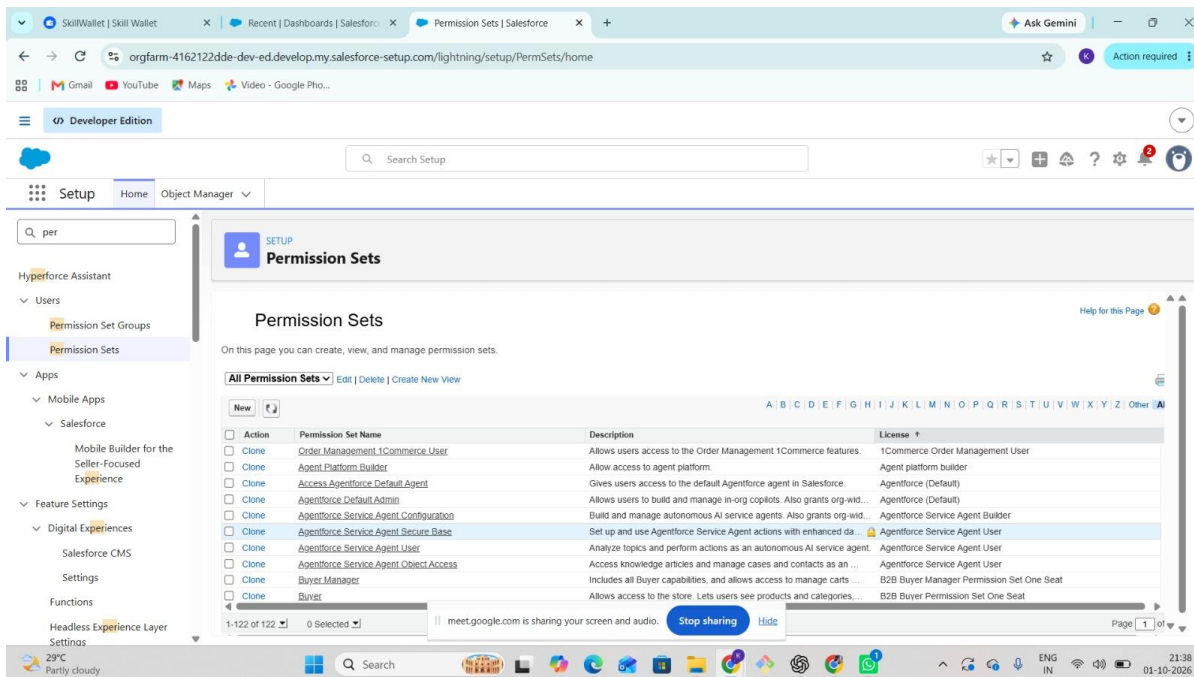
Prompt template: Retrieve Parcel Details. API Name: Retrieve_Parcel_Details. The prompt retrieves Parcel__c information where Parcel_ID__c matches user input and returns Parcel Name, Parcel ID, Status, Weight and Estimated Delivery information.



Salesforce Prompt Builder – Retrieve Parcel Details template.

Milestone 6 – Flows

The reference implementation creates an Auto-Launched Flow. It accepts an input variable for the parcel ID, retrieves the Parcel record, invokes the parcel-details action, assigns the response output, and returns the result. The supplied implementation screenshot shows the Get Records, Decision and Assignment structure.

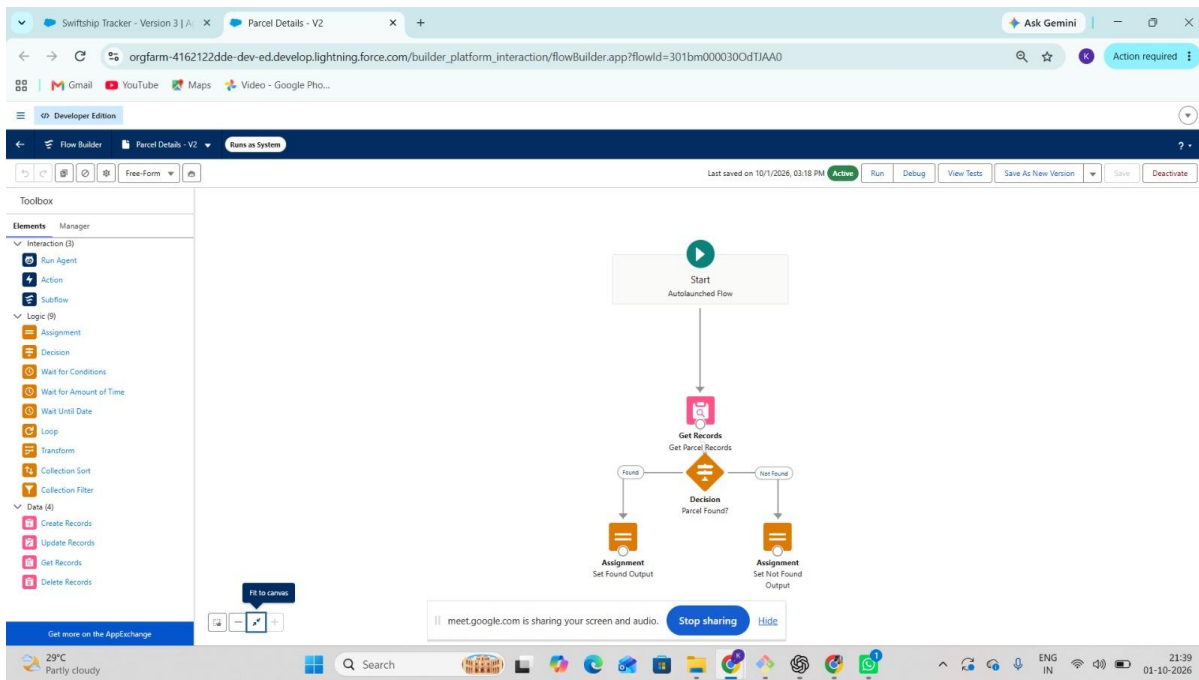


Salesforce Flow Builder – Parcel Details flow with Get Records, Decision and Assignments.

6.3 UI/UX Development & Customization

This phase focuses on secure access control and the AI-driven AgentForce experience.

A SwiftShip permission set is created and object-level and field-level access is granted for Parcel__c, Delivery__c, Sender__c and Receiver__c.



Salesforce Permission Sets – AgentForce-related permission sets visible.

Milestone 2 – Assigning Permissions to Agent User

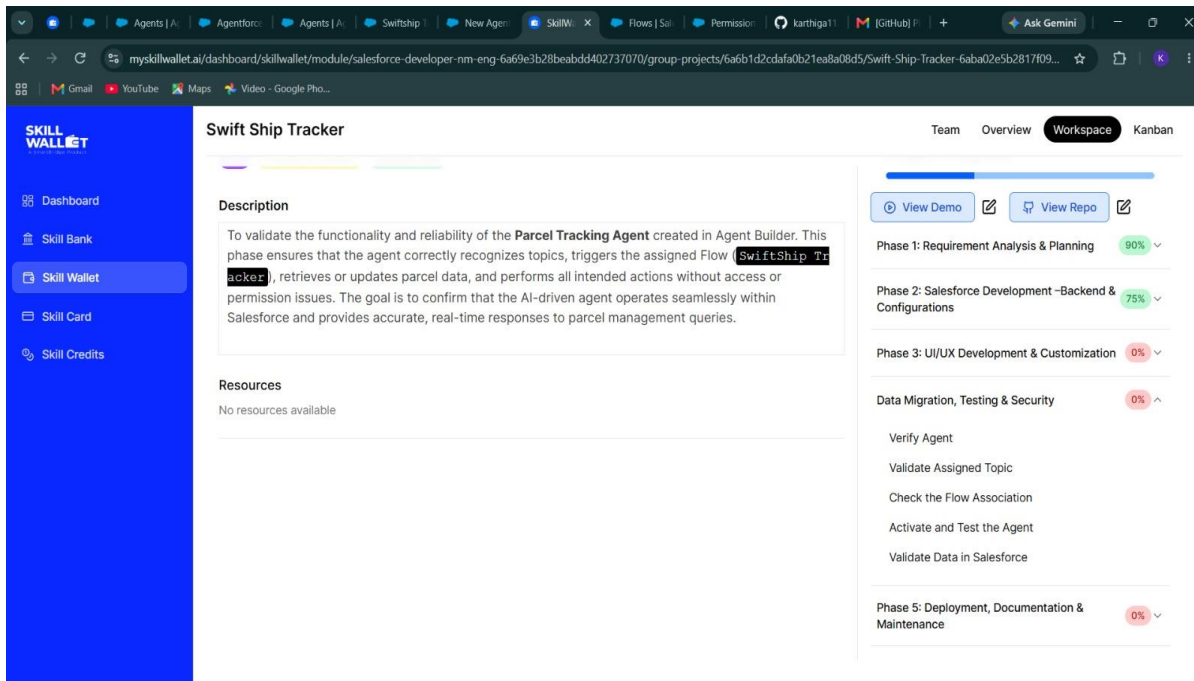
The SwiftShip permission set is assigned to the appropriate Agent / Einstein Service Agent user.

Milestone 3 – Org Setup for AgentForce

- Enable Salesforce Default Agent
- Enable Einstein Setup
- Enable Agentforce services
- Open Agentforce Agents and Agent Builder

Milestone 4 – Building the Agent

- Create the parcel-tracking subagent/topic.
- Add the Flow as an Agent Action.
- Configure the Parcel Updates subagent.
- Rename the agent to SwiftShip Tracker.
- Save and activate the agent.



Salesforce Agentforce Studio – project environment.

Milestone 5 – Assigning Permissions to the Agent User

Verify that the Agent user has the required permission set and access to the parcel-related objects and fields.

6.4 Data Migration, Testing & Security

Verify Agent

Go to Setup → Agentforce Agents and confirm that the Parcel Tracking Agent is available.

Validate Assigned Topic

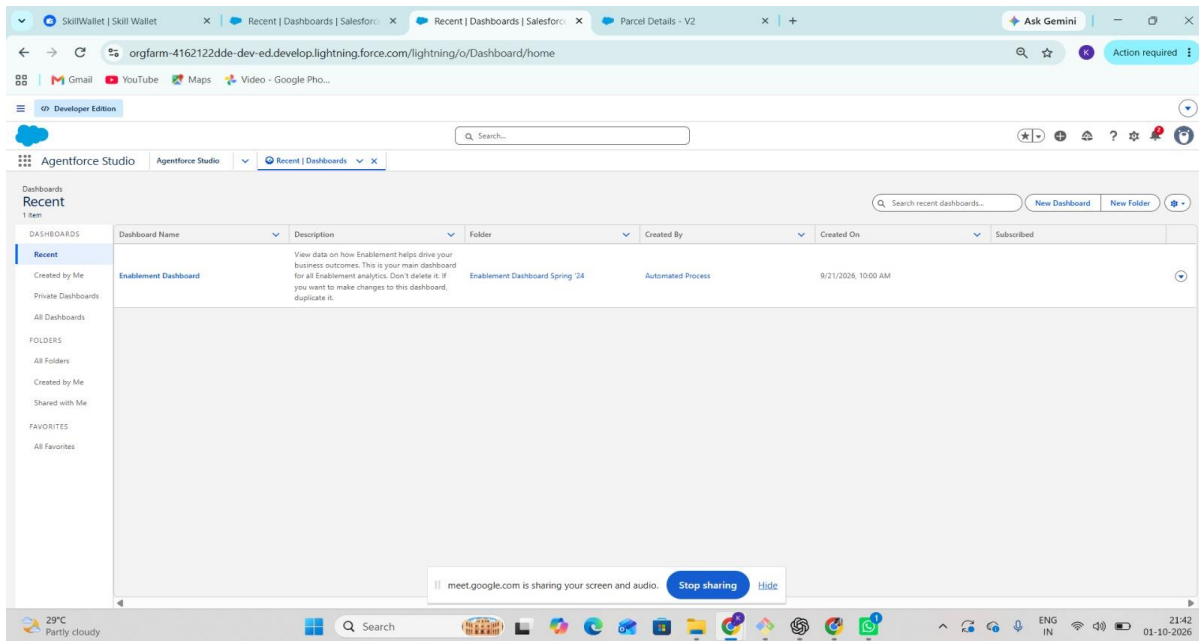
In Agent Builder, confirm the Parcel Tracking / SwiftShip Tracker topic and verify that the SwiftShip Tracker Flow is linked to the topic.

Check the Flow Association

Go to Setup → Flows, locate the relevant flow, verify it is Activated, and confirm that it handles Parcel ID input, parcel details retrieval and the required confirmation/status response.

Activate and Test the Agent

Use Preview / Try It Out with sample queries such as “Track parcel ID P12345” or “Update delivery status for parcel P67890 to Delivered.”



SkillWallet – Parcel Tracking Agent validation/testing task.

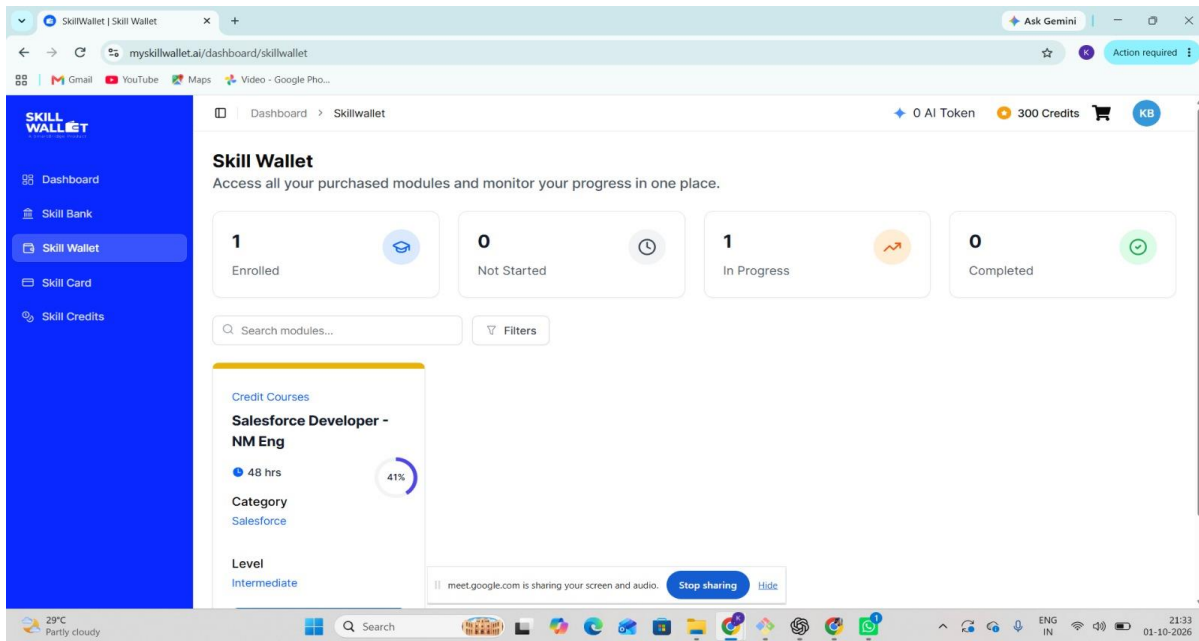
Validate Data in Salesforce

After the agent runs the Flow, open Parcels (Parcel__c), search for the tested Parcel ID, and verify that the status or related fields are updated correctly and that any Flow-created records appear as expected.

6.5 Deployment, Documentation & Maintenance

The reference describes this as an academic/prototype implementation within a Salesforce Developer Org. Production UAT, sandbox staging, and enterprise CI/CD are outside the stated scope.

- Monitor parcel, customer and delivery data.
- Detect and resolve failed Flow executions and configuration errors.
- Refine AgentForce topics and prompts.
- Review permission and access configurations.
- Improve AI response quality and user interactions.
- Maintain project documentation and demo material.

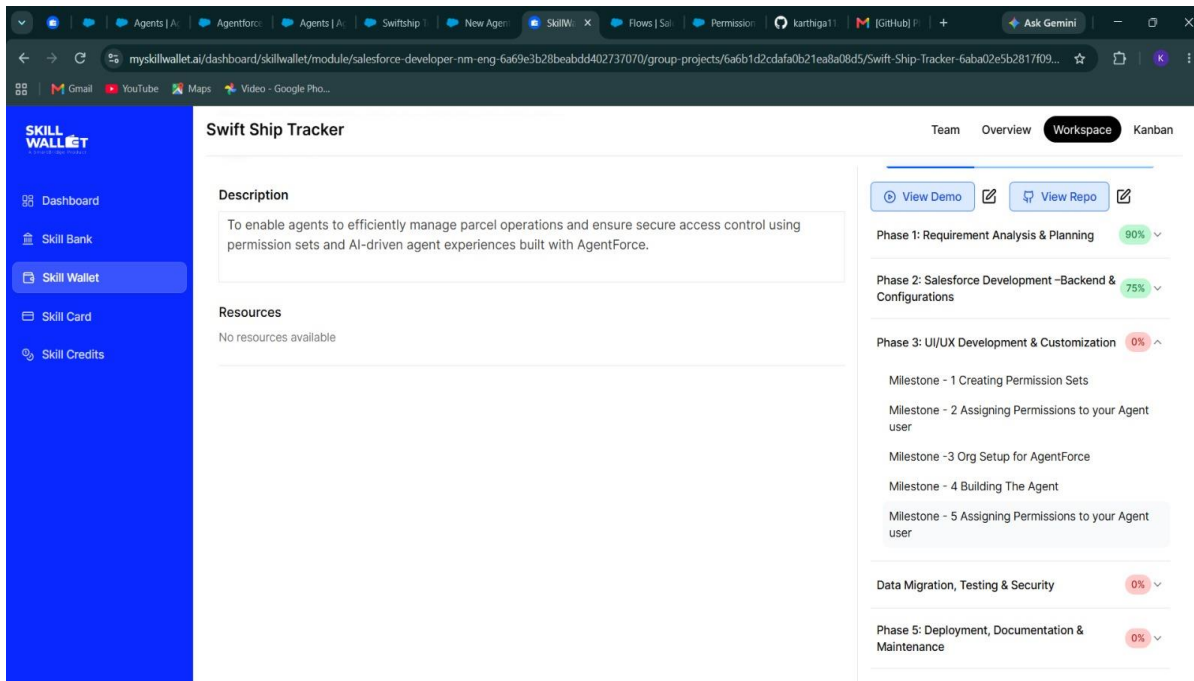


SkillWallet – Phase 3 milestones: Permission Sets and AgentForce configuration.

7. FUNCTIONAL & PERFORMANCE TESTING

Testing validates object availability, prompt retrieval, Flow execution, AgentForce topic association, permissions, agent activation, and final Salesforce data.

Test ID	Test Case	Expected Result	Evidence
TC-01	Parcel object	Parcel__c exists and is accessible	Salesforce Object Manager screenshot
TC-02	Delivery object	Delivery__c exists and is accessible	Salesforce Object Manager screenshot
TC-03	Prompt Builder	Parcel details can be retrieved from Parcel ID	Prompt Builder screenshot
TC-04	Flow	Get Records and decision/assignment logic executes	Flow Builder screenshot
TC-05	Permission Set	Agent receives required object access	Permission Sets screenshot
TC-06	Agent	Agent/topic/Flow configuration is available	Agentforce screenshot / validation task
TC-07	Data validation	Parcel record reflects tested operation	Salesforce record validation



SkillWallet – Skill Wallet course/project progress.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized parcel and customer information.
- Improved shipment visibility.
- Automation of logistics processes through Salesforce Flow.
- Conversational parcel tracking using AgentForce AI.
- Prompt Builder support for self-service information retrieval.
- Reduced manual effort.
- Timely delivery notifications.
- Role-based security and controlled access.
- Reports and dashboards for operational monitoring.
- Scalable cloud-based architecture.

Disadvantages

- Dependence on Salesforce configuration and availability.
- AgentForce response quality depends on prompt and data configuration.
- External logistics integrations can add complexity.
- Flows and prompts require maintenance when business rules change.
- Users require suitable Salesforce / Experience Cloud access.
- Initial configuration can require significant effort.
- Production-scale deployment requires additional testing and release management.

9. CONCLUSION

The SwiftShip Tracker demonstrates how Salesforce can be used to build a centralized parcel management and tracking solution. The project integrates custom objects, relationships, Flows, Apex, Email Alerts, Experience Cloud, AgentForce AI, Prompt Builder, security controls, and reporting to support parcel booking, delivery tracking, status updates, and customer communication.

The integration of AgentForce AI with Prompt Builder and Salesforce Flow provides a conversational approach to parcel tracking, enabling users to obtain shipment information through an AI-driven experience while the underlying Salesforce records remain the system of record.

10. FUTURE SCOPE

- Enhance real-time delivery tracking.
- Expand Experience Cloud customer self-service.
- Add advanced delivery-performance analytics.
- Integrate external logistics and GPS systems.
- Improve AgentForce prompts and topic coverage.
- Introduce enterprise sandbox and CI/CD deployment.
- Add more automated parcel operations and notifications.
- Scale the solution for multi-location logistics operations.

11. APPENDIX

A. Custom Objects

Parcel__c, Delivery__c, Sender__c, Receiver__c

B. Standard Objects

Account, Contact, Case, Task, EmailMessage, User

C. Automation

Record-Triggered Flows, Auto-Launched Flows, Scheduled Flows, Approval Processes, Email Alerts

D. Apex

Apex Classes and Batch Apex

E. AI

AgentForce AI and Prompt Builder

F. UI

Lightning App Page, Lightning Record Pages, Experience Cloud

G. Security

Profiles, Role Hierarchy, Permission Sets, Sharing Rules, Field-Level Security

H. Analytics

Reports and Dashboards

I. Implementation Evidence

The following screenshots supplied by the student are embedded in this report at the corresponding implementation stages:

- SkillWallet – Swift Ship Tracker workspace showing project phases and progress.
- SkillWallet – Parcel Tracking Agent validation/testing task.
- SkillWallet – Phase 5 deployment, documentation and maintenance scope.
- SkillWallet – Phase 2 milestones: Salesforce Account, Objects, Tabs, Fields & Relationships, Prompt Builder and Flows.
- SkillWallet – Phase 3 milestones: Permission Sets and AgentForce configuration.
- SkillWallet – Skill Wallet course/project progress.
- Salesforce Object Manager – Parcel custom object.
- Salesforce Object Manager – Delivery custom object.
- Salesforce Permission Sets – AgentForce-related permission sets visible.
- Salesforce Flow Builder – Parcel Details flow with Get Records, Decision and Assignments.
- Salesforce Prompt Builder – Retrieve Parcel Details template.
- Salesforce Agentforce Studio – project environment.